MODULE 2 SUMMATIVE ASSESSMENT

Technology Evaluation, Literature Review, and Conceptual Framework

Organization	FlowBikes
Emerging Technology	Artificial Intelligence (AI) / Generative AI – AI Concierge
Proposed Research Topic	Evaluating the Use of an AI Concierge for Customer Inquiry Handling and Product Recommendation at FlowBikes

PART A – TECHNOLOGY EVALUATION MATRIX

Evaluation Criterion	Analysis / Evidence
Organizational Need	AI directly addresses FlowBikes' reliance on unorganized messaging for customer technical inquiries. An AI Concierge can automate common bicycle-related questions and reduce repetitive manual responses.
Technical Suitability	AI can be integrated into the proposed web platform and connected to relevant product and inventory information, making it technically aligned with the existing system concept.
Feasibility	The project proposal identifies accessible technologies such as Supabase, Vercel, and Groq. The main concerns are AI/API integration and free-tier limits, which can be managed through prioritization and usage controls.
Business / Organizational Value	The AI Concierge can provide faster responses, improve access to bicycle information, reduce repetitive workload, and support product recommendations.
Security	Risks include exposed API credentials, unauthorized access, malicious prompts, and inappropriate data access. API keys, database permissions, input validation, and access controls should be protected.
Privacy	Customer inquiries may contain personal information. The system should minimize unnecessary data collection and protect information sent to external AI services.
Ethics	The AI should be transparent about being AI, avoid misleading recommendations, and use human oversight when questions require staff judgment.
Governance	FlowBikes should define who manages the AI, what data it may access, how responses are monitored, and how

	inaccurate responses are corrected.
Sustainability	AI consumes computing resources. Efficient prompts, request limits, and responsible use of cloud resources can reduce unnecessary computation.
Overall Evaluation	AI Concierge technology is relevant and potentially feasible for FlowBikes because it directly addresses customer inquiry handling. Its use should be supported by security, privacy, accuracy, human oversight, and governance controls.

Part A Summary

The evaluation indicates that AI Concierge technology fits the FlowBikes problem because it can automate routine customer inquiries and support product discovery. However, accuracy, privacy, security, customer trust, and governance must be considered during implementation.

PART B – LITERATURE REVIEW MATRIX

Author/Year	Research Focus	Technology/Method	Key Findings	Relevance to Our Research
Nicolescu & Tudorache (2022)	AI chatbots in customer service	Systematic literature review	Customer experience is influenced by chatbot, customer, and contextual factors.	Supports considering usability, expectations, and context for the FlowBikes AI Concierge.
Olujimi & Ade-Ibijola (2023)	Automating customer queries	NLP; systematic review	NLP can automate customer inquiries and support customer-service operations.	Directly supports conversational AI for FlowBikes' inquiry problem.
Mariani, Hashemi & Wirtz (2023)	AI-powered conversational agents	Systematic literature review	Research includes trust, NLP, communication, and business value.	Supports examining both benefits and adoption concerns.
Kang, Lee et al. (2023)	Customer-service chatbot satisfaction and loyalty	Empirical survey	Conversational and service-recovery quality influence satisfaction.	Provides direction for evaluating customer interaction quality.
Retail chatbot study (2025)	Consumer adoption of GenAI retail chatbots	Consumer survey	GenAI improved perceived usefulness and human-likeness,	Relevant to balancing AI usefulness and privacy in retail.

			while privacy concerns also increased.	

1. Common Findings or Themes

The literature commonly indicates that AI chatbots can automate customer-service inquiries and reduce the need for human intervention in routine questions. It also shows that chatbot quality, responsiveness, usability, trust, privacy, and customer expectations can influence user experience and acceptance.

2. Research Gap

Much of the literature examines general customer-service environments or larger retail contexts. There is less direct evidence concerning an AI Concierge for a small bicycle retailer with specialized product inquiries and live inventory information. This creates a gap concerning technical bicycle inquiries, product recommendations, customer experience, and current inventory information in a specialized small-retail context.

3. Connection to Our Research

The reviewed literature supports the proposed research by showing that conversational AI can automate inquiries and influence customer experience. For FlowBikes, these findings provide direction for evaluating whether an AI Concierge can assist customers with bicycle specifications, product questions, and recommendations while maintaining privacy, security, and human oversight.

PART C – CONCEPTUAL FRAMEWORK

ORGANIZATIONAL PROBLEM

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Manual spreadsheet inventory records
Unorganized customer messaging
Delayed technical inquiries
Difficulty providing product information

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PROPOSED EMERGING TECHNOLOGY
AI / Generative AI – AI Concierge

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TECHNOLOGY / APPLICATION FUNCTIONS

Automated customer inquiries
Bicycle specification assistance
Product recommendations
Relevant product-information retrieval
Direct routing to seller when needed

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EXPECTED ORGANIZATIONAL OUTCOMES

Faster customer responses
Better access to product information
Reduced repetitive manual workload
Improved customer experience
More organized inquiry handling

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CONTEXTUAL FACTORS

User acceptance • Organizational readiness • Data accuracy • Technical skills
Cybersecurity • Privacy • Governance • Cost / API limitations

Conceptual Framework Explanation

The conceptual framework proposes that the FlowBikes organizational problem of manual inventory records, unorganized messaging, and delayed customer inquiries can be addressed through an AI-powered Concierge integrated into the web-based system. The AI Concierge can provide automated responses to common bicycle-related questions, assist customers in understanding product specifications, and provide product recommendations using available system information. When an inquiry requires human assistance, the system can direct the customer to the seller. These functions are expected to improve response time, reduce repetitive manual work, improve access to product information, and support a more organized customer-service process. Successful implementation depends on user acceptance, organizational readiness, technical skills, accurate product data, cybersecurity, privacy, governance, and API or infrastructure limitations.

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